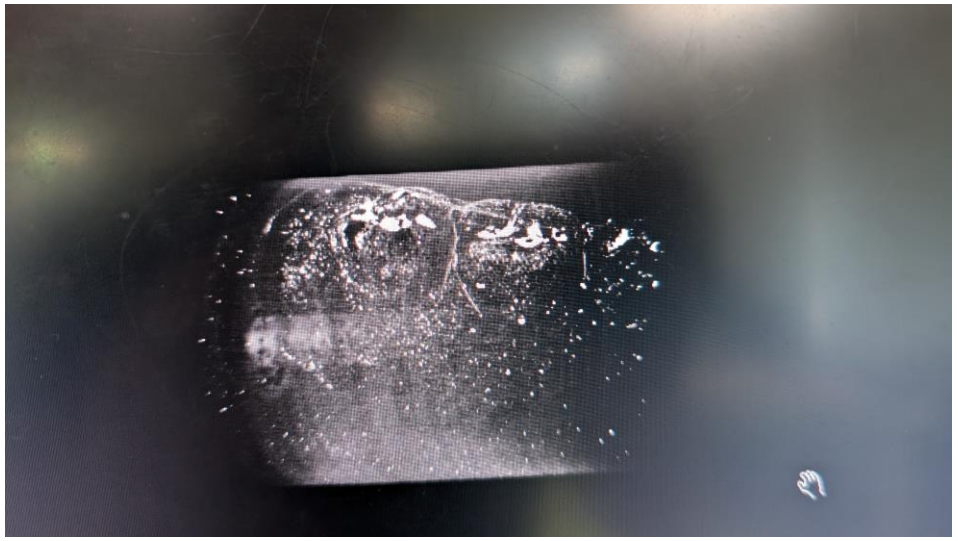
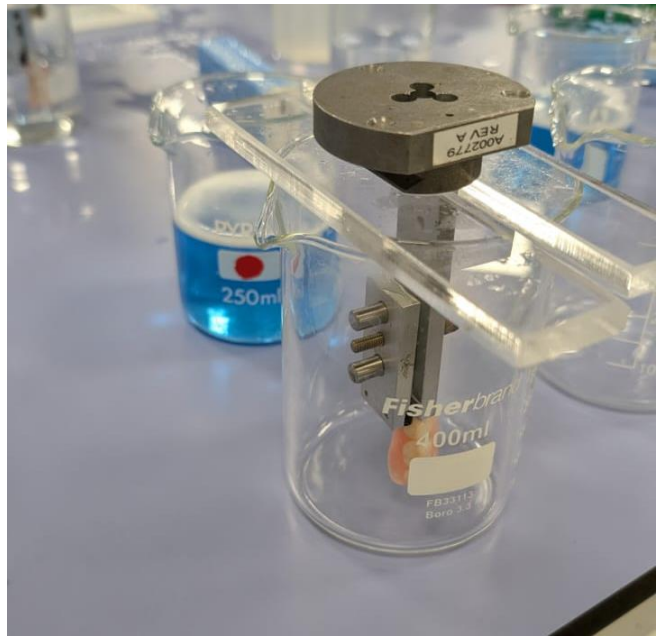
Design Portfolio

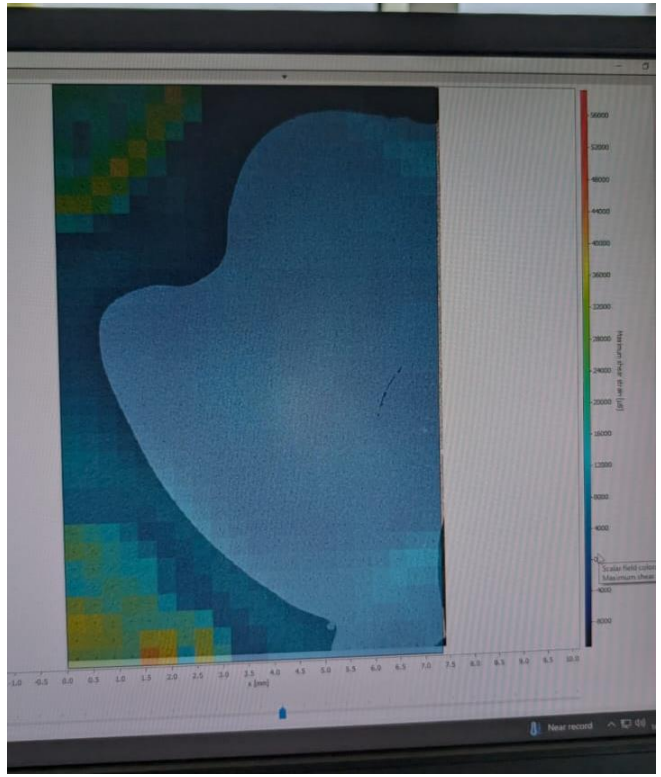
- SHIVA RAMKUMAR

Surface Abrasion Test Project between University of Portsmouth and Haleon



Grew Biofilm on Dental Implant

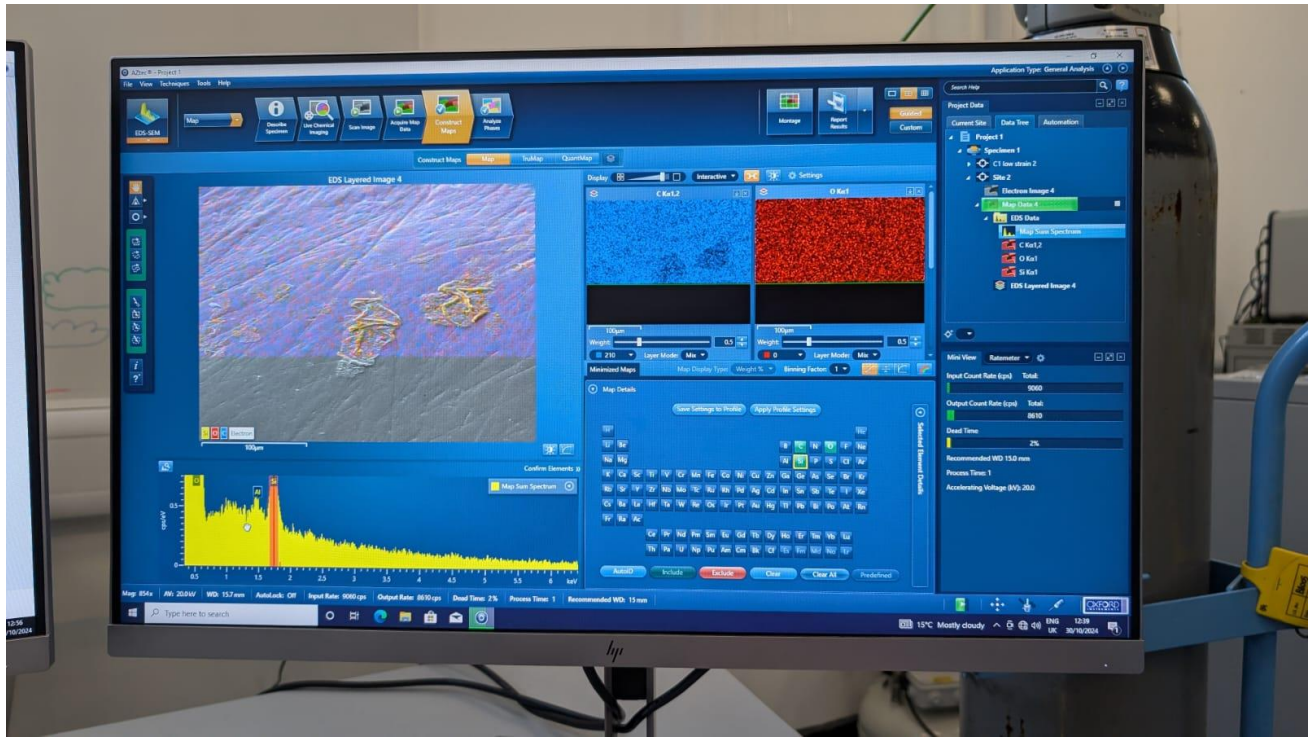




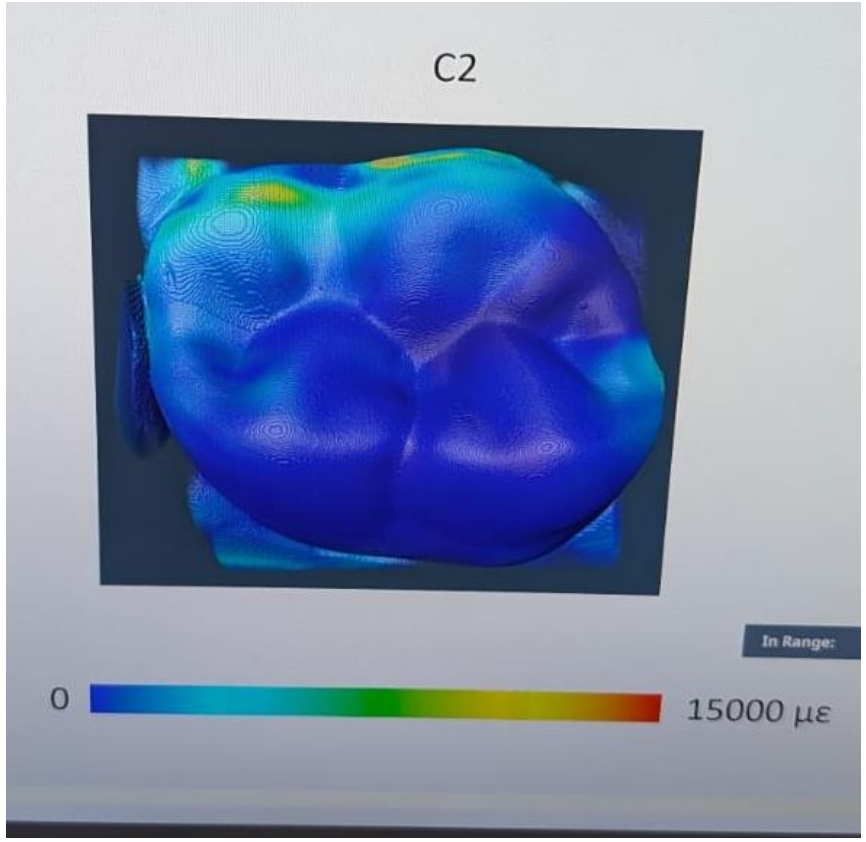
Used Digital Volume Corelation to calculate Strain on Pre vs Post XCT Scans

During my role as a Research Associate at the University of Portsmouth

Correlated abrasion region using SEM

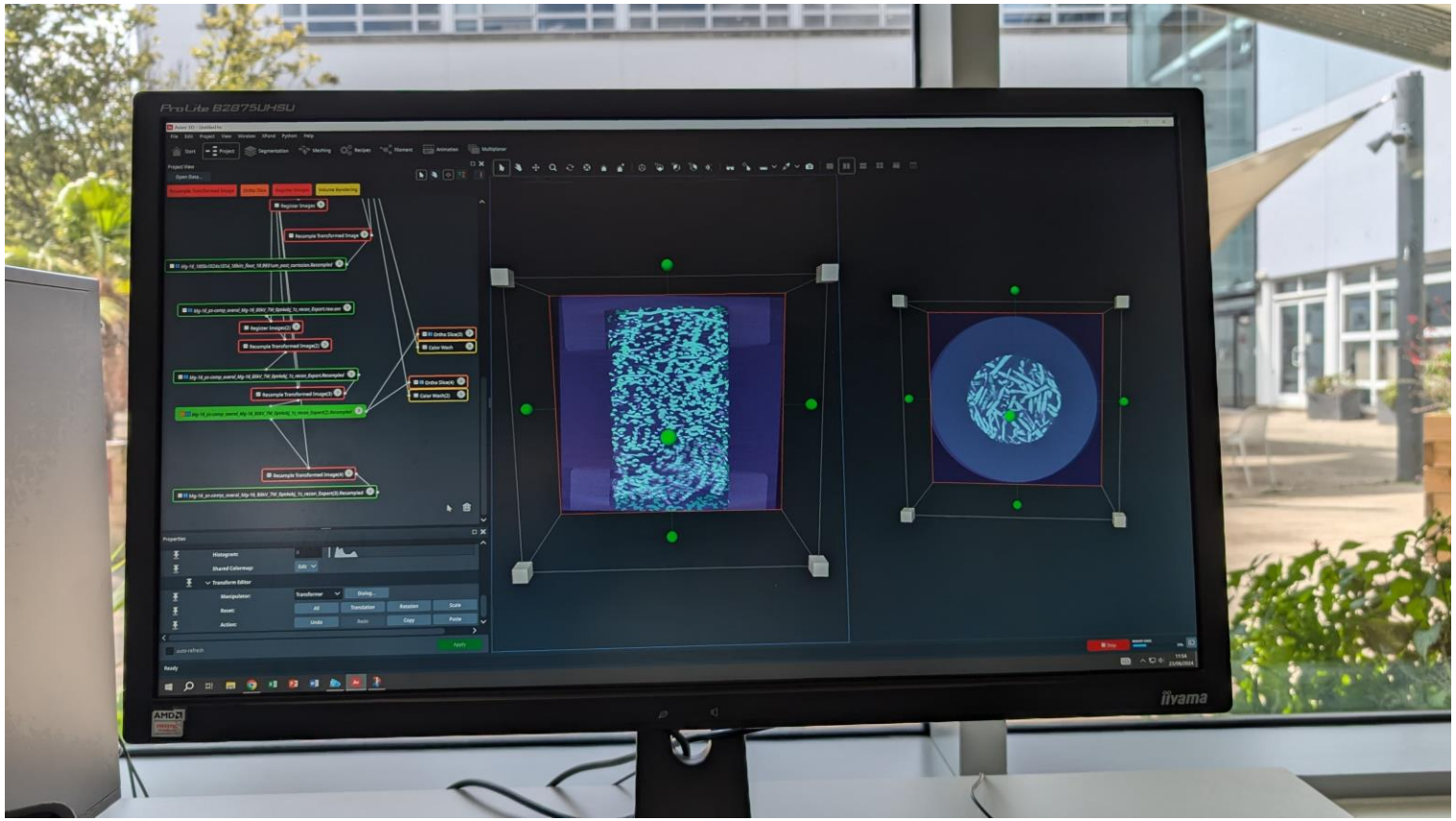


Applied strain map using Avizo



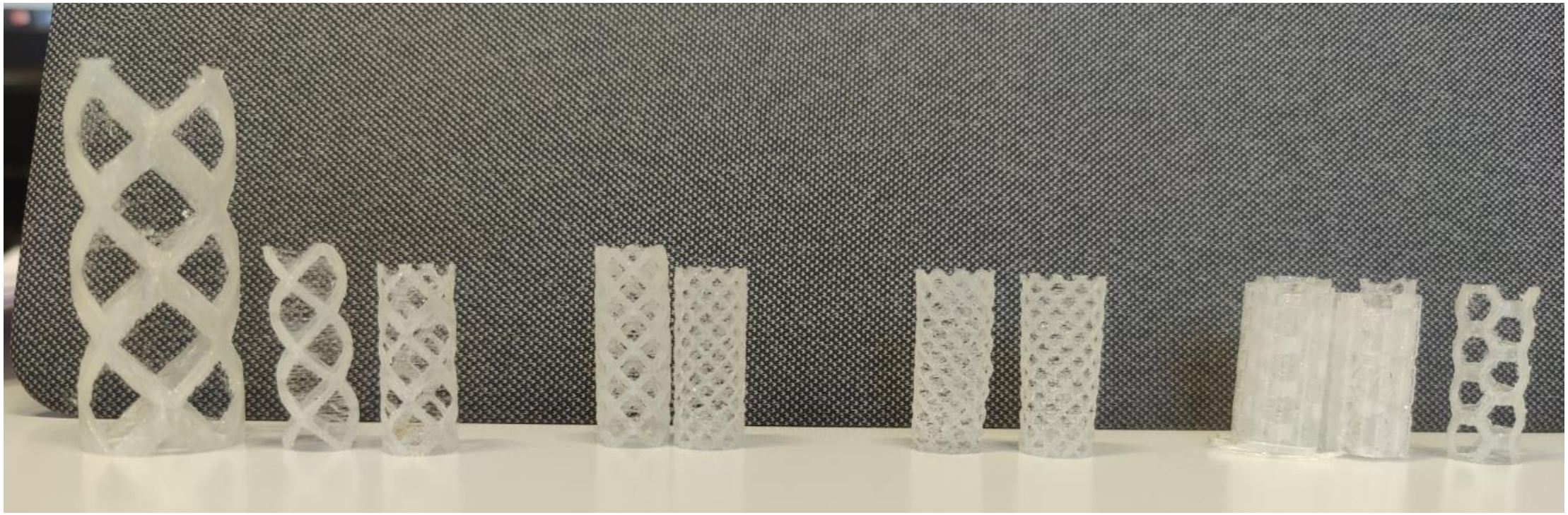
Magnesium implants under compression XCT Scan study for regenerative bone support using AVIZO

During my role as a Research Associate at the University of Portsmouth

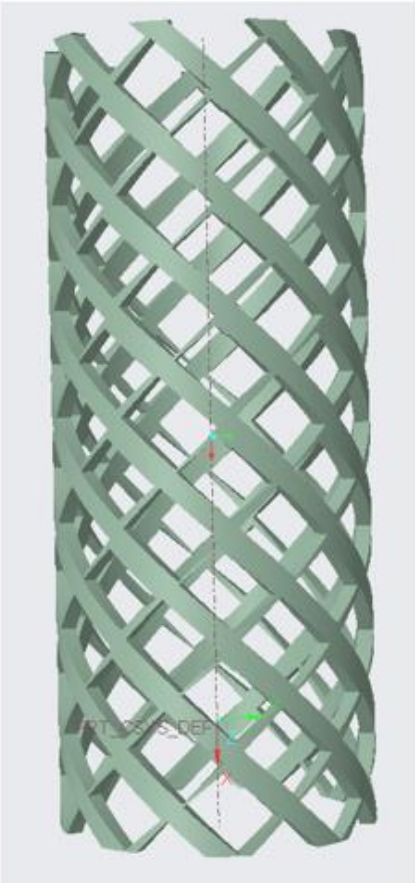


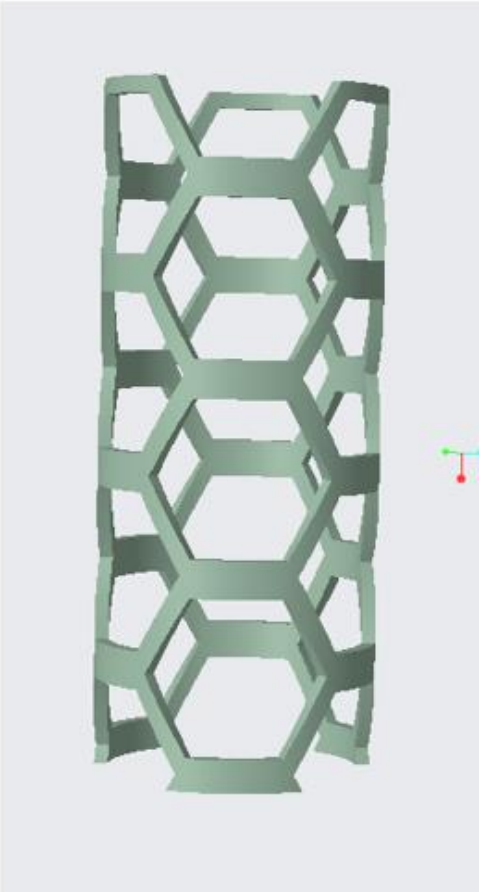
4D printing of Vascular stents using PLA

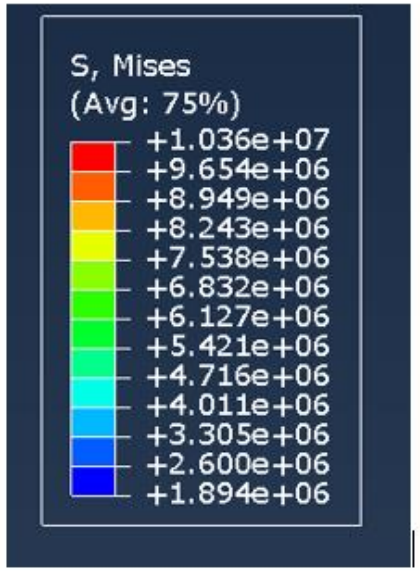
My Individual Project during my Masters in Mechanical Engineering



Designs using CAD for expansion upon external stimulus on 4D Material

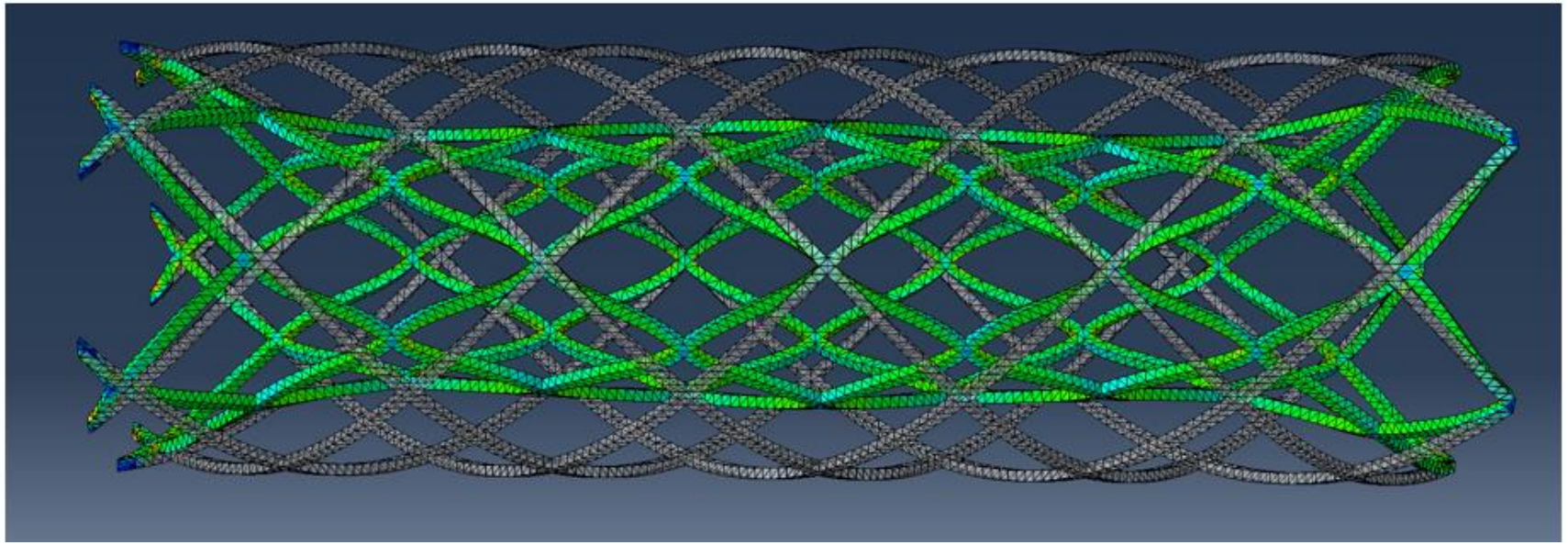




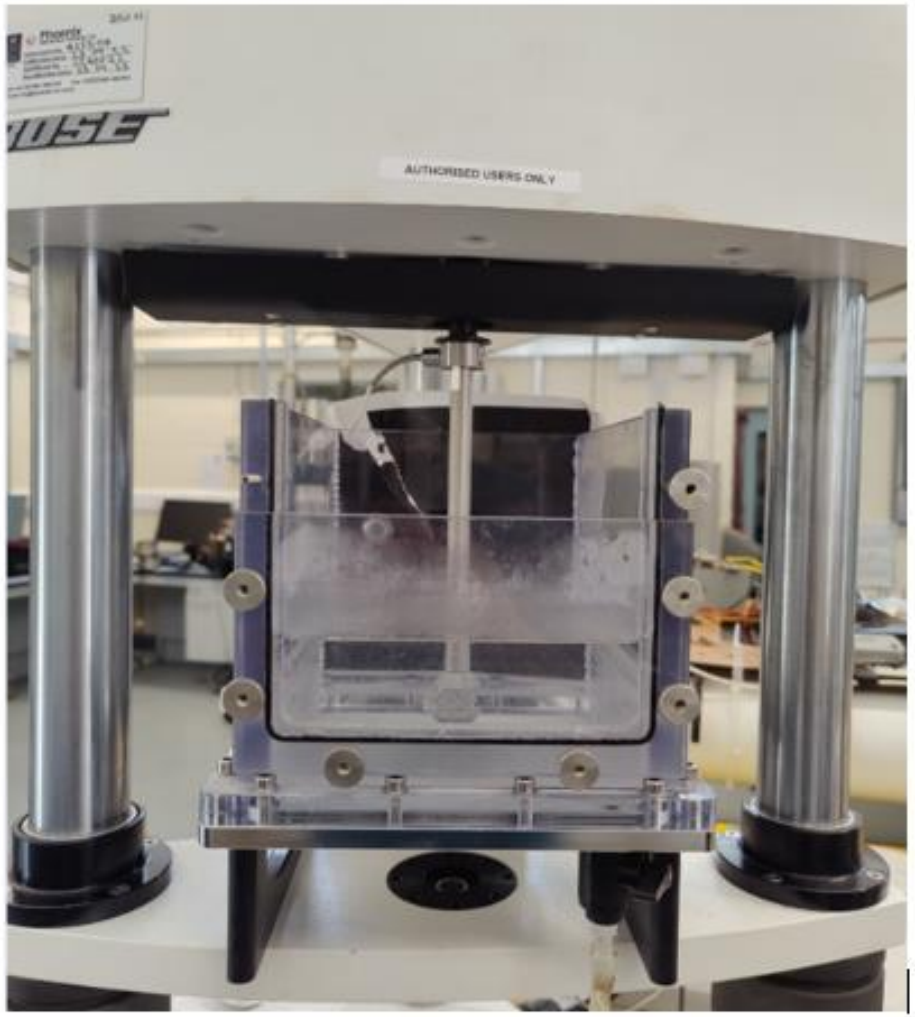


FEA ANALYSIS

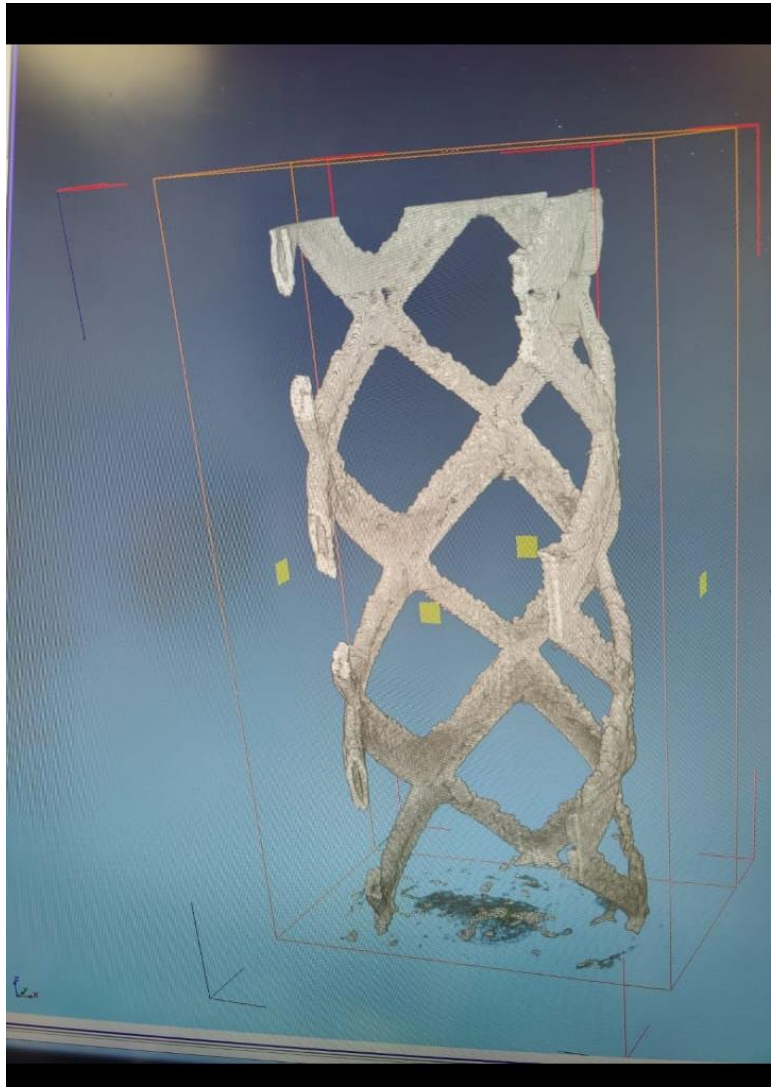
Calculated Stress and Strain using Simulations in Abaqus FEA Software by applying suitable Load



Performed Mechanical Tests on the 3D printed specimen for checking Structural Integrity



Compression Test



Porosity Test

CT Scans

Stent A

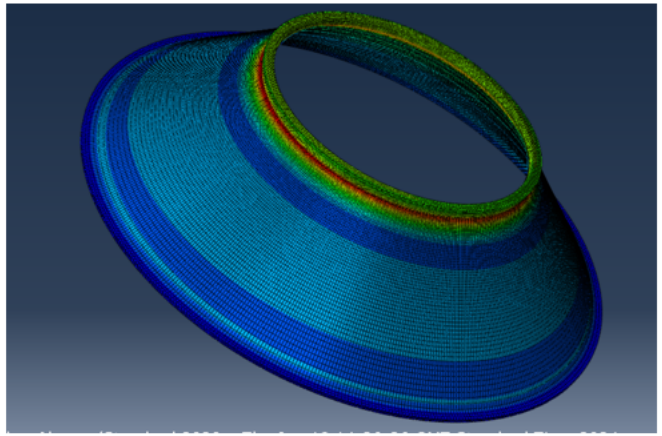
Stent B

Stent C

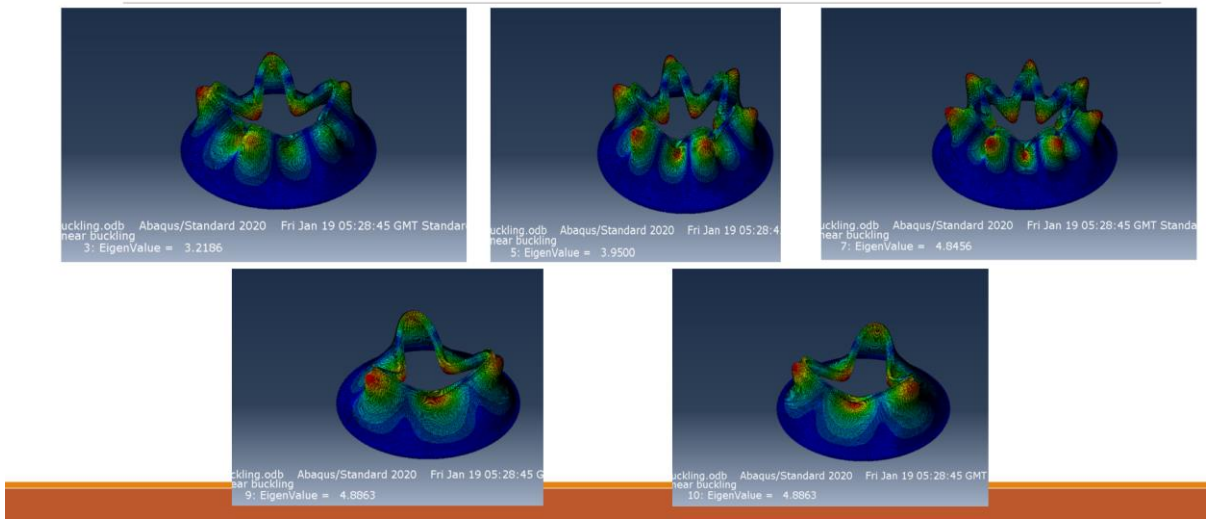
FINITE ELEMENT ANALYSIS

Payload Adapter System

(PAS 937) FOR SOYUZ LAUNCHER

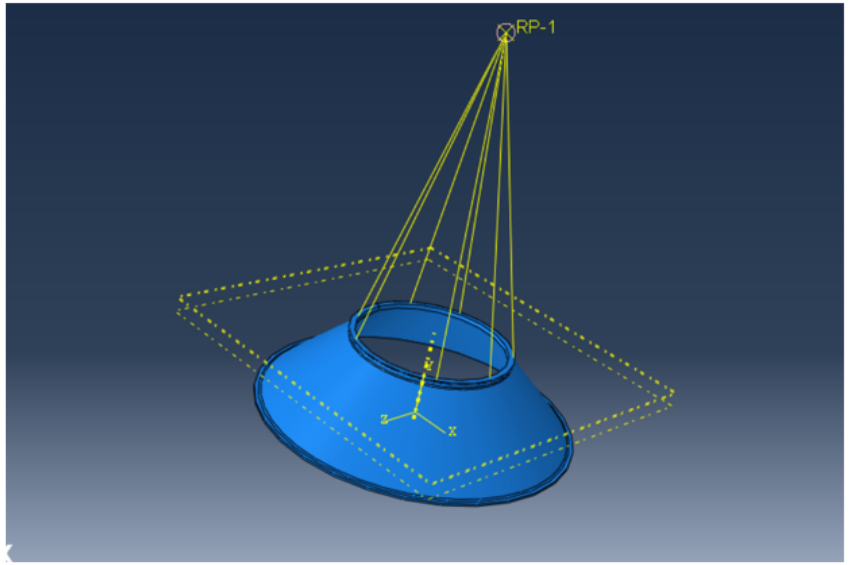


Linear Buckling Analysis



My FEA Coursework during my Masters

Load applied as Concentrated Force using a reference point

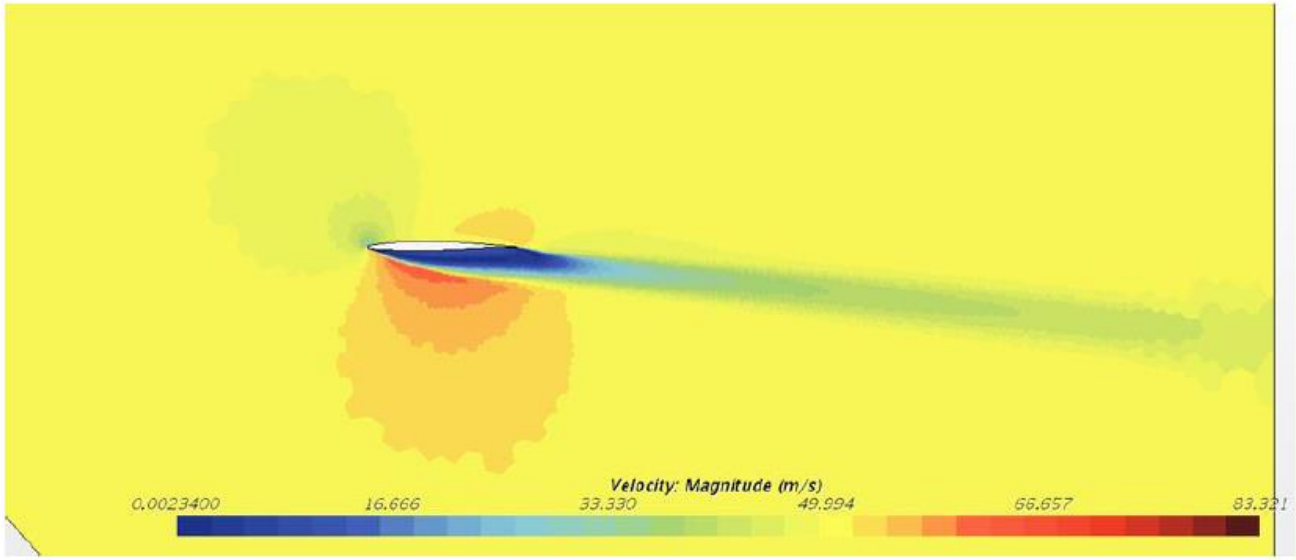


The PAS 937 S is designed and qualified to support a payload of 3500 kg centered at 1.8 m from the separation plane. (from the Soyuz manual)

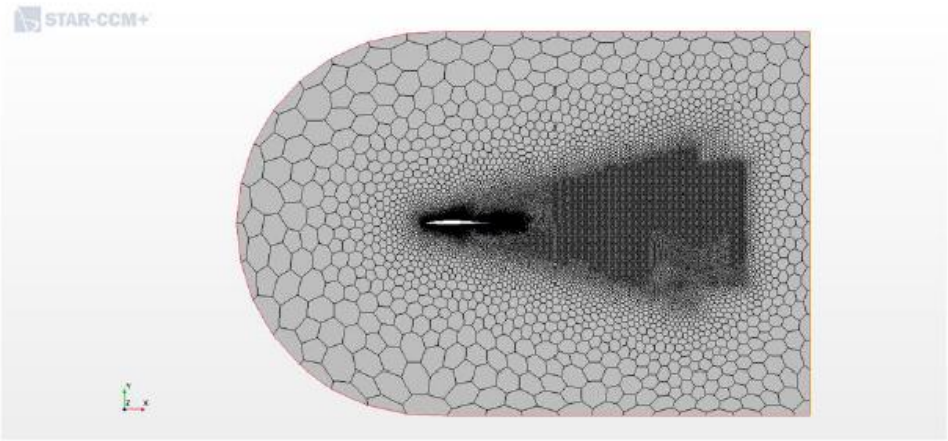
Load in axial direction under maximum acceleration (3500*9.81*4.3)/0.00974 = 1515816.2217659

Shear force experienced when the Rocket is travelling at an angle of 45 degrees 0.4g =13734

AIFOIL SIMULATION FOR CFD COURSEWORK USING STAR CCM+ TO UNDERSTAND FLOW SEPARATION

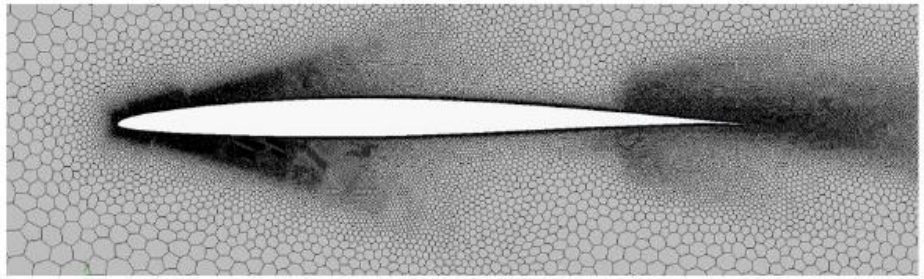


Mesh Convergence



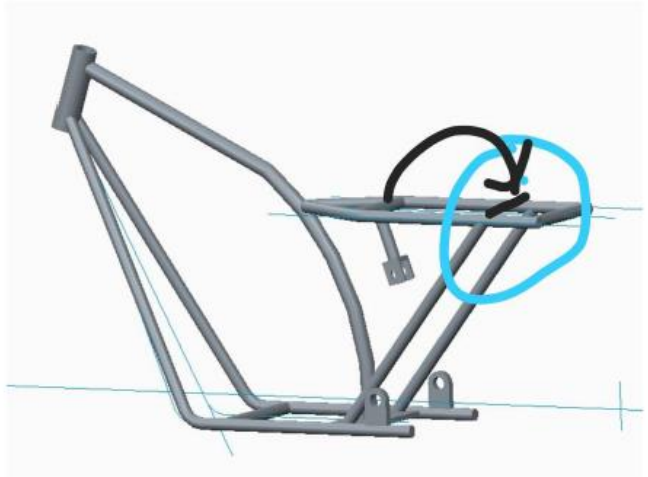
NACA 64 206 MESH 1

NACA 64 206 MESH



My CFD Coursework during my Masters in Mechanical Engineering

DESIGNED ELECTRIC CAFÉ RACER FOR AMT MOTOCORP 2019 ELECTRIC VEHICLE COMPETITION



During my Undergraduate degree in India